GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

3. A drawing has a 50° angle. A scale copy is made at k = 2. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 25°
- ○ C. 50°
- ○ D. 52°

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 5 cm^2. Its scale copy has area 80 cm^2. Find k, and explain why k is not 16. *1 pt*

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 11 in? *1 pt*

10. 1 cm = 0.39 in. How many inches is 90 cm? *1 pt*

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 5 yd? *1 pt*

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

13. 1 gal = 3.785 L. How many litres is 9 gal? *1 pt*

14. 1 ft = 12 in. A tile is 7 ft^2 in area. How many square inches is that? *1 pt*

- ○ A. 1,008 in^2
- ○ B. 84 in^2
- ○ C. 144 in^2
- ○ D. 168 in^2

15. Rate on 5 Nov: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

b. How many US dollars is 54 Canadian dollars? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

b. It starts a 1,500-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

b. With 8 cups on hand, how many servings can be made? *1 pt*

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second colour. How many gallons is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A 450-km drive. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how long? Explain. *1 pt*